

```
In [8]: import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler
        from sklearn.neighbors import KNeighborsClassifier
        from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [9]: df = pd.read_csv(r"C:\Users\MERIN\IRIS.csv")
        df
```

```
Out[9]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

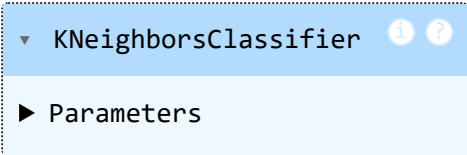
150 rows × 5 columns

```
In [10]: X = df[['sepal_length', 'sepal_width', 'petal_length', 'petal_width']]
        Y = df['species']
```

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In [11]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
```

```
In [12]: scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
In [13]: k = 5
knn = KNeighborsClassifier(n_neighbors=k)
knn.fit(X_train, Y_train)
```

```
Out[13]: 
▼ KNeighborsClassifier ⓘ ?
▶ Parameters
```

```
In [15]: Y_pred = knn.predict(X_test)
```

```
In [16]: print("Accuracy:", accuracy_score(Y_test, Y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(Y_test, Y_pred))
```

Accuracy: 1.0

Confusion Matrix:

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```